



Operational field monitoring of interactive vortex-induced vibrations between two parallel cable-stayed bridges[☆]



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ABSTRACT

An in-depth study was conducted on interference VIV between two parallel cable-stayed bridges with respect to the mutual motion of both decks downstream as well as upstream. The mechanical damping ratios of both bridges were estimated by the Natural Excitation Technique (NExT) combined with the Eigen Realization Algorithm (ERA) method. The test setup in a wind tunnel takes two wind directions as well as the identified damping ratios of both decks into consideration. The findings, based on parametric wind tunnel tests, suggest that interference VIV is possible, even in the downstream area of the bridge although this has not been reported before. However, the higher lock-in velocity as well as the higher damping ratio of the downstream bridge would be expected to decrease the possibility of VIV. The interactive behavior was further examined using field monitoring data and the results were in good agreement with the findings obtained in wind tunnel tests, in terms of the threshold wind velocity, the frequency components of the motion and the amplitude ratio between the two bridges. Unfortunately, however, a strong wind was not observed opposite to the main wind direction and it was not possible to confirm the interactive behavior for this situation.

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1. Introduction

As the volume of traffic increases, the necessity of building new bridges in the form of a parallel bridge has become a necessity. The Tacoma Narrow Bridge, the Puente Juan Pablo Duarte (Larsen et al., 2000) and the Haihe Bridge (Meng et al., 2011) are examples of recently constructed parallel cable-supported bridges. The Jindo Bridge (Fig. 1) is one of the parallel bridges adopting cable-stayed structural systems. The First Jindo Bridge was built in 1984 and the Second Jindo Bridge was opened to traffic in 2005. For convenience, the First Jindo Bridge is hereafter referred to as Bridge 1 and the Second Jindo Bridge as Bridge 2. Vortex-induced vibrations (VIVs) have been reported a few times in the case of Bridge 2, since it has opened to traffic.

Interference effects between two parallel cable-stayed bridges was investigated by Kimura et al. (2008), who reported governing factors through wind tunnel tests. Meng et al. (2011) proposed a mitigating measure of the interference effect for two adjacent cable-stayed bridges. It was concluded that gap distance as well as deck shape were dominant parameters that affect the interference effect in both studies. However, the more intensive studies were also recommended, in order to expand our general insights related to this complicated interactive behavior.

On April 19, 2011, a significant VIV with a duration of one hour was detected in Bridge 2, as shown in Fig. 2. The single amplitude of acceleration exceeded the recommended value of 0.5 m/s² (Korean Society of Civil Engineers (KSCE), 2006) and double amplitude of displacement reached 0.4 m at the center of the main span in Bridge 2 located upstream. Since the noticeable vibration in the upstream deck was an interesting phenomenon, a series of wind tunnel tests were performed in an attempt to identify the main sources of VIV (Seo et al., 2013). The study pointed out that the amplitude of the VIV in Bridge 2 was sensitive to the mechanical damping of the bridge. The estimated damping ratio of Bridge 2 was lower than the design value and this was proposed to constitute one of the sources of the VIV. It was also

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Fig. 1. The parallel twin Jindo Bridge.

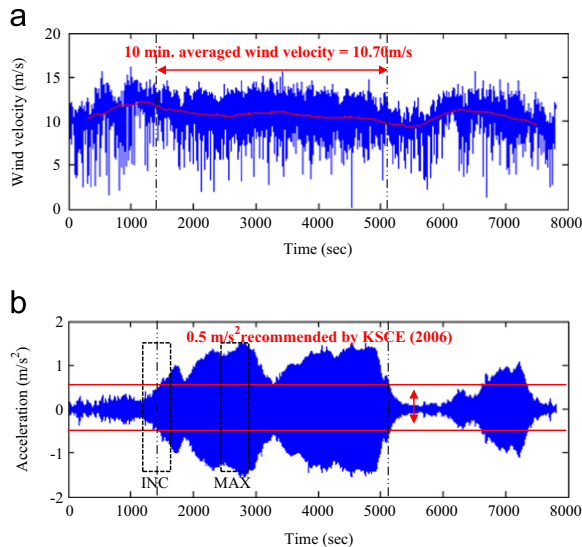


Fig. 2. Observed wind velocity and interference VIV at the center of the main span in Bridge 2: (a) wind velocity and (b) acceleration.

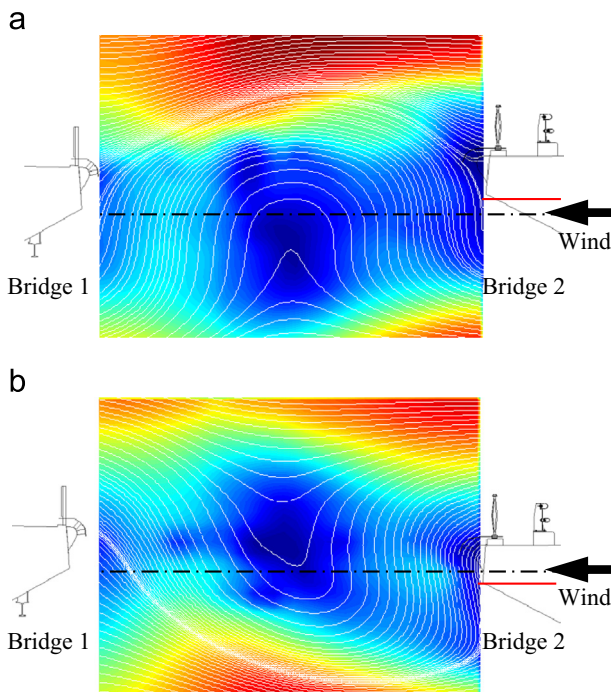


Fig. 3. Alternating “Ω” and “U” shaped large vortices between two decks during the significant VIV in the Bridge 2 at the moment the deck reaches: (a) top position and (b) bottom position.

found that large vortices were developed between two decks with alternating “Ω” and “U” shaped streamlines, as shown in Fig. 3. Seo et al. (2013) traced the pattern of the streamlines and flow

speeds between the two decks and successfully demonstrated an interference effect on the upstream deck. However, the study mainly focused on the motion of the upstream deck which was subjected to a huge VIV.

Fig. 3 shows that the large vortex streams developed behind Bridge 2 upstream and pass through Bridge 1 downstream. Consequently, it is possible that the vortex streams could affect the motion of, not only Bridge 2, but Bridge 1 as well even though Bridge 1 unaffected by the huge VIV in 2011. At this moment, several questions can be raised. Is the interference VIV only observable in Bridge 2? Otherwise, what are the conditions required for the interference VIV for Bridge 1 to be observed and why have VIVs not been reported yet for Bridge 1? In addition, VIVs in Bridge 2 have been reported only a few times in spring seasons after the opening of the bridge. What conditions need to be satisfied for observing a significant VIV in Bridge 2?

According to Seo et al. (2013), several factors may be involved in answering these questions, including the main wind direction and the differences between the inherent damping ratios and the triggering wind velocities for the two bridges. Accordingly, more in-depth analyses should be conducted to clarify the interactive behaviors between bridges by taking these potential parameters into consideration.

Based on these research backgrounds, parametric wind tunnel tests were carried out for the two scaled parallel decks mounted on spring supports. The order of disposition of the two decks can be changed, as a function of the wind direction. The mechanical damping ratios were set to the values identified from an operational modal analysis. The interactive behaviors were also investigated with field data obtained from the built-in monitoring system of two bridges. The monitoring data covers a wide range of excitations in both bridges including a typhoon as well as daily winds. The overall field observations were in good agreement with the result of wind tunnel tests and demonstrated the interesting interactive behaviors between two bridges.

2. Identification of damping ratios from an operational modal analysis

2.1. Built-in sensors and three-day monitoring of vibration

Since the amplitude of the interference VIV in the upstream bridge was found to be sensitive to the mechanical damping ratio (Seo et al., 2013), the identification of the actual structural damping ratio of two bridges would be a starting point of investigation on interactive behavior. Acceleration data accumulated for three days between 2012/10/15 10:00 and 2012/10/17 24:00 were utilized for the operational modal analysis. During the period, the bridges were subjected to ambient vibrations cause by the movement of traffic as well as ambient, daily winds.

The vertical accelerations of bridge deck were monitored at the center of the main spans in both bridges. The vertical responses of the decks were obtained by averaging two vertical accelerations at both sides of the cross-section. Unfortunately, one of the accelerometers in Bridge 1 was not functioning at the time of measurement, and the vertical acceleration at the other side of the cross section is only utilized by ignoring the contribution of the torsional motion of the deck.

Fig. 4 shows the built-in sensors utilized in this study. Acceleration was measured in gal (cm/s^2) with a sampling frequency of 100 Hz. Wind direction and wind velocity were also recorded by ultrasonic anemometers that were installed on Bridge 2 and Bridge 1 at a height of 3 m from the top of decks, respectively. Even though both anemometers were connected to the same data

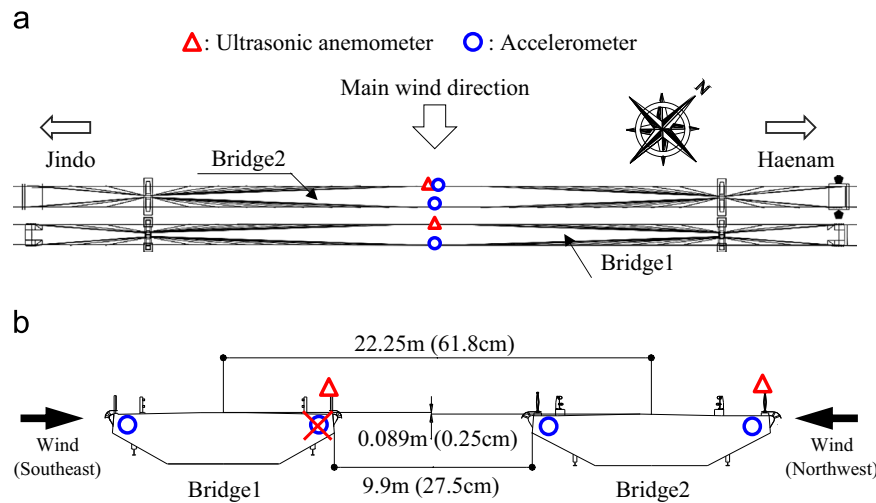


Fig. 4. Built-in monitoring sensors used for the observation of interference VIV: (a) plane view and (b) cross section.

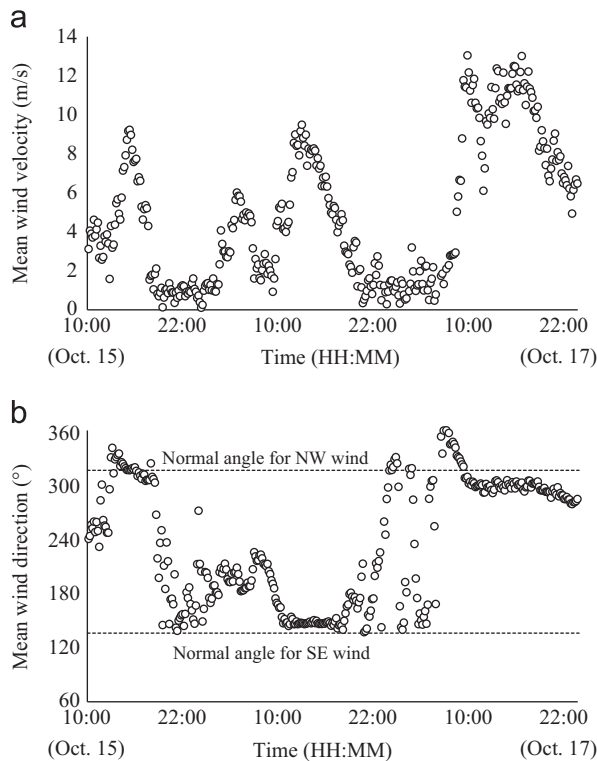


Fig. 5. Wind properties for a three-day monitoring period: (a) 10-min average wind velocity and (b) 10-min average wind direction.

logger scanning 100 Hz, the actual capacities of the sampling rate of the anemometers were 1 Hz and 10 Hz for Bridge 2 and Bridge 1, respectively. The entire body of three-day raw data was divided into 10-min data and averaged over the time, as shown in Fig. 5. The wind data were measured by the anemometer located upstream of Bridge 2. In wind tunnel test measurements, the wind velocity at the anemometer position increased by about 10% compared to the free upstream velocity due to the local interference effect by the deck itself. The 10-min average wind velocity was less than 13 m/s and the wind direction varied between Northwest (NW) and Southeast (SE). However, most of the high winds were normal to the bridge decks, as shown in Fig. 5(a) and (b).

2.2. Evaluation of damping ratios for both bridges

For evaluating the damping ratios of both bridges, the three-day acceleration responses were again divided into 20-min long data. Since wind velocity was low, the contribution of aerodynamic damping was assumed to be negligible in estimating the mechanical damping ratios. In fact, the maximum aerodynamic damping ratio of the first vertical mode was found to be somewhat less than 0.1% for the wind velocity ranges shown in Fig. 5 based on the single-mode flutter analysis theory (Simiu and Scanlan, 1996) with experimentally obtained flutter derivatives for the bridge sections. The Natural Excitation Technique (NExT) (James et al., 1993) combined with the Eigen Realization Algorithm (ERA) (Juang and Phan, 2001; Siringoringo and Fujino, 2008) was applied to each 20-min data set.

The auto correlation function or equivalently impulse response function in NExT was derived from an inverse Fast Fourier transform (FFT) of the auto power spectral density (PSD) function of the vertical acceleration (Bendat and Piersol, 2000), as shown in Fig. 6. The sample acceleration in Fig. 6(a) is for a low wind velocity of 4 m/s and mainly induced by vehicles that are crossing over the span. The amplitude of vibration exceeds 5 gal and the maximum value reaches 25 gal depending on the vehicle type. One excited signal lasts for 20 to 40 s which corresponds to the crossing time over the main span. The PSD of acceleration in Fig. 6(b) demonstrates the excitation of several modes by crossing vehicles. The auto correlation function in Fig. 6(c) was obtained by the inverse FFT with the number of data for 2^{16} .

The modal order in the ERA was chosen based on the number of peaks that are visible in PSD diagrams. Even though a modal order equivalent to fifteen modes was employed in obtaining the dynamic properties of bridges, only the first one corresponding to the first vertical mode was considered. The size of the Hankel matrix varies between 500×500 and 1700×1700 for the 100 Hz sampled auto correlation function. The damping ratio was evaluated for the case in which the maximum modal participation factor is produced among different sizes of the Hankel matrix.

The identified damping ratios are shown in Figs. 7 and 8 for Bridge 2 and Bridge 1, respectively. The damping ratios of Bridge 2 were scattered between 0.1% and 0.6% with a mean value of 0.29%. This is much lower than the recommended damping ratio of 0.4% for the wind-resistant design recommendations for a steel cable-stayed bridges (KSCE, 2006) and point out one of the sources of notable vibration in Bridge 2 in 2011 (Seo et al., 2013). Here, the

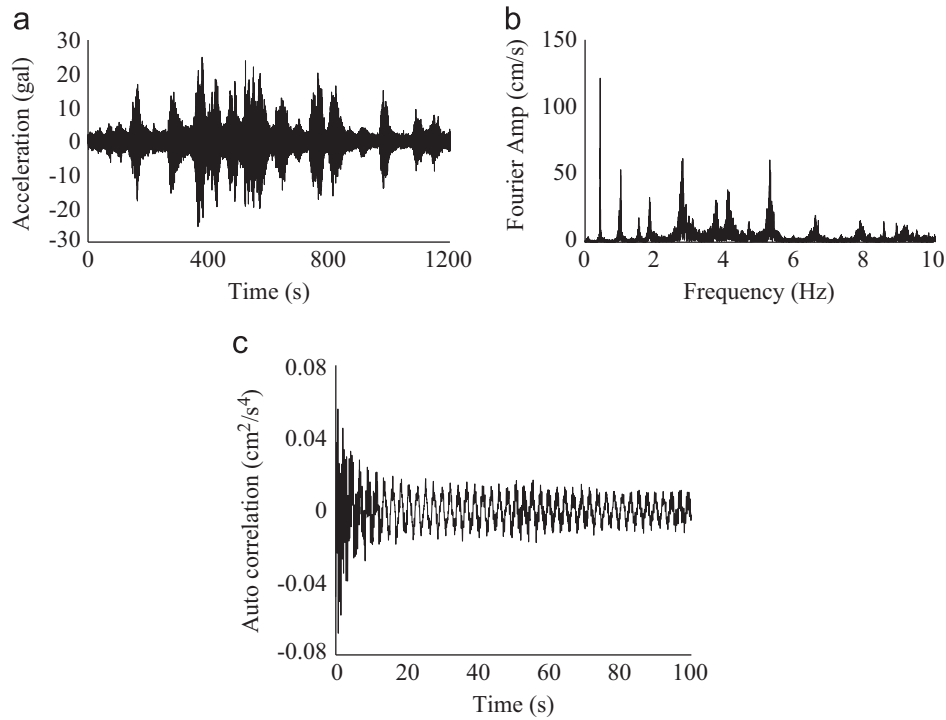


Fig. 6. Measured vertical acceleration at the center of the main span of Bridge 2 during 2012/10/15 13:40 to 14:00: (a) time history for 20-min, (b) Fourier spectrum and (c) auto correlation function or equivalently impulse response function.

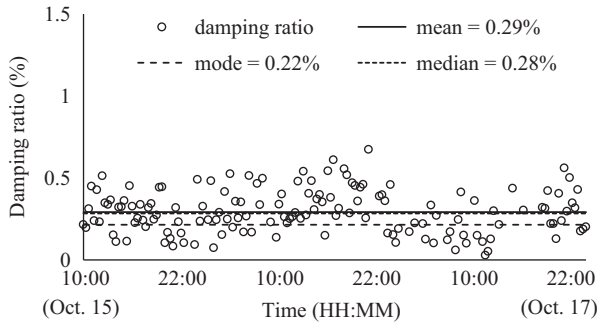


Fig. 7. Estimated damping ratio for the first vertical mode of Bridge 2.

damping ratio is represented as a percent relative to the critical damping ratio.

Bridge 1 was closed to traffic during the monitoring period and the source of vibration was only the wind. Fig. 8(a) shows the estimated damping ratios for all 20-min data sets, while Fig. 8(b) presents the damping ratios only for the data in which the component of the first vertical mode is dominating. The estimated damping ratio is higher than that of Bridge 2 with resulting in a mean value of 0.54–0.63% depending on the data considered.

3. Interactive behavior between two decks in a wind tunnel

3.1. Setup of wind tunnel tests

The interactive behavior between the two bridges was investigated in a wind tunnel operated by the Department of Civil and Environmental Engineering at Seoul National University, Korea. The test section of the wind tunnel is 1.5 m in height, 1.0 m in width and 4.0 m in length. The section models for both bridge decks were prepared with a length scale of $\lambda_L = L_m/L_p = 1/36$, where L_m and L_p represent the length of the model and prototype, respectively. The section models are basically identical to those

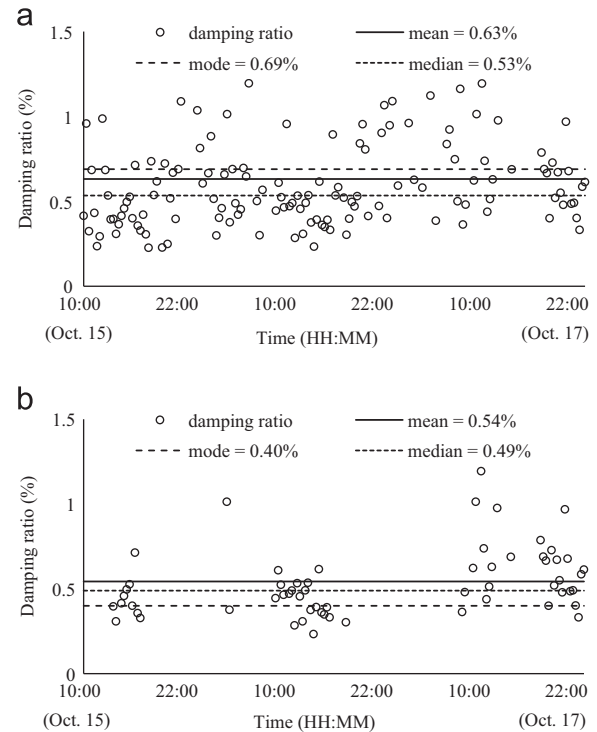


Fig. 8. Estimated damping ratios for the first vertical mode of Bridge 1: (a) overall data and (b) only considering the first-vertical-mode dominant data.

used in the preceding study (Seo et al., 2013). However, in order to monitor the interference phenomena in a higher wind velocity, the frequency scale increases from 5 to $\lambda_f = f_m/f_p = 1/\lambda_T = 11.3$ where λ_T is the time scale and f_m and f_p represent the frequency of the model and prototype, respectively. Table 1 shows the setup parameters for both section models, which were on spring-supported systems. The damping ratios for the first vertical modes were set to the actual damping ratios (0.28% for Bridge 2 and 0.58%

Table 1
Similarity scales and setup parameters for the sectional models

Parameters	Similarity scale	Bridge 2			Bridge 1		
		Prototype	Model (target)	Model (measured)	Prototype	Model (target)	Model (measured)
Length (m)	$\lambda_L = 1/36$	32.400	0.900	0.900	32.400	0.900	0.900
Width (m)	λ_L	12.690	0.353	0.353	11.860	0.329	0.329
Height (m)	λ_L	2.750	0.076	0.076	2.800	0.078	0.078
Mass (kg/m)	λ_L^2	8978	6.927	6.904	6950	5.363	5.376
1st vertical frequency (Hz)	$\lambda_f = 11.3$	0.436	4.919	4.919	0.513	5.780	5.920
Vertical damping ratio (%)	1	0.280	0.280	0.280	0.580	0.580	0.550

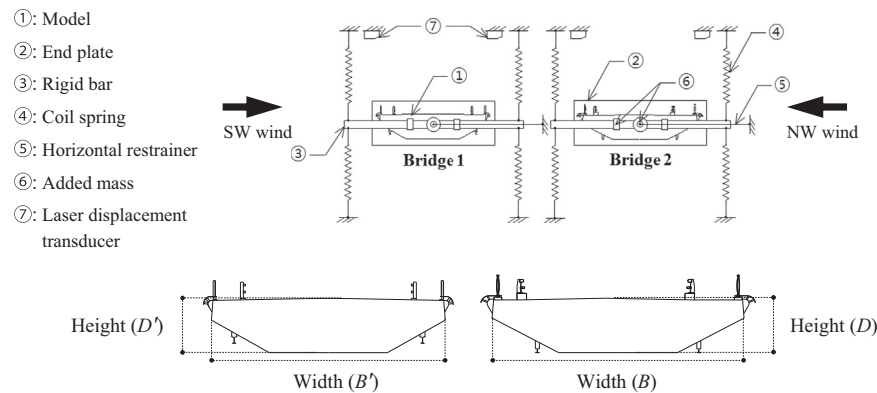


Fig. 9. Setup of wind tunnel tests and nomenclature.

for Bridge 1) shown in Figs. 7 and 8. To better observe the VIV, however, the damping ratios were also set to a minimal level of 0.14% for both bridges. This level of damping is hereafter referred to as the low damping ratio. The sensor deployments and the spring-supported section models are shown in Fig. 9. The turbulence intensity was maintained at less than 1% throughout the wind tunnel tests. All the parameters related to wind tunnel tests are hereafter represented in a prototype scale for purposes of easier comparison with the field observations.

3.2. Strouhal numbers in parallel disposition

Prior to the vibration tests, the Strouhal numbers were measured so as to estimate the lock-in velocity. The Strouhal number for a NW wind, in which Bridge 2 was located upstream, was 0.109 while that for a SE wind, in which Bridge 1 was located upstream, was found to be 0.128. The shedding frequency of the upstream bridge was measured between the two decks and the Strouhal number was normalized to the height of the upstream deck. For more information, the Strouhal numbers for the single deck of Bridge 2 and Bridge 1 were determined to be 0.123 and 0.145, respectively. Fig. 10 shows the change in vortex shedding frequencies as a function of wind velocity for the case of a NW wind. The shedding frequencies for the spring-supported setup clearly show two lock-in plateaus which demonstrate two VIVs with different frequencies tuned to each deck.

3.3. Interactive behavior for NW wind

The dominant wind direction for the bridges is NW. Bridge 2 is located upstream for this wind direction. Fig. 11(a) shows the single amplitude–velocity (A – V) curves for this wind direction with the setup of low damping ratios. Fig. 11 represents double axes to demonstrate the results of the wind tunnel test in a non-dimensional scale as well as a prototype scale. The non-dimensional parameters are normalized with respect to Bridge 2. Since the first vertical frequency of Bridge 2 ($=0.436$ Hz) was

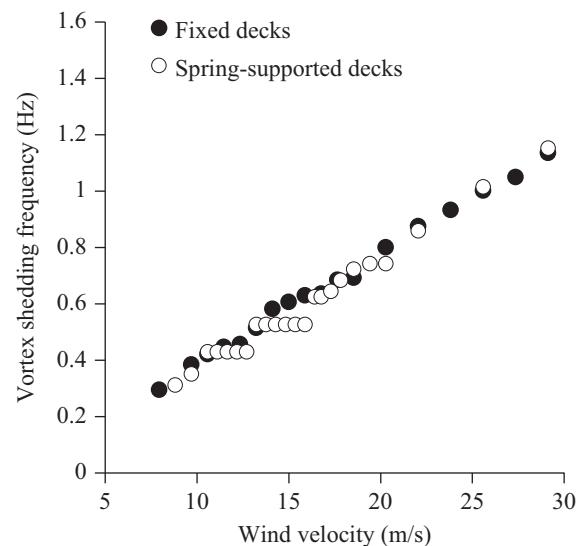


Fig. 10. Vortex shedding frequencies measured between two decks when Bridge 2 locates upstream for NW wind.

lower than that of Bridge 1 ($=0.513$ Hz), the VIV starts first in Bridge 2 at a wind velocity of 10.9 m/s. With a further increase in wind velocity, the VIV moves to Bridge 1 at a wind velocity of 12.9 m/s. The observed VIV velocities are consistent with the measured Strouhal number. The single amplitude of the displacements reaches 0.24 m in Bridge 2 and 0.32 m in Bridge 1 in terms of the root mean square (RMS) value in a steady state motion. Although significant VIVs were mainly reported for Bridge 2 up to the present time, it is interesting that Bridge 1 was possibly subjected to an even larger VIV than Bridge 2 if the damping ratios were same in both bridges.

Fig. 11(b) also shows the A – V curves for the setup with actual damping ratios. The VIV velocities are unchanged while the amplitudes of vibration decreased to an extent in both bridges,

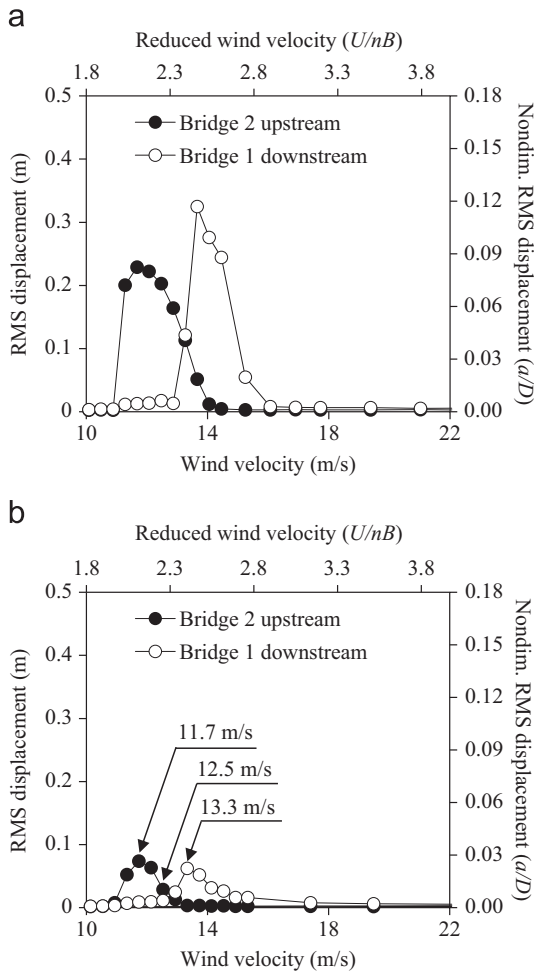


Fig. 11. A–V curves for NW wind the setup of (a) low damping ratios and (b) actual damping ratios.

as expected. However, the maximum amplitude of vibration in Bridge 2 still reaches 0.07 m in RMS displacement. The allowable limit of vibration recommended in the design guidelines (KSCE, 2006) is 0.066 m in terms of maximum displacement (Seo et al., 2013), which is equivalent to 0.047 m in terms of RMS displacement based on a harmonic steady-state oscillation. The amplitude of displacement in Bridge 1 further decreases owing to a relatively high damping ratio and now become slightly less than that of Bridge 2. Considering the first vertical frequency, the allowable limit of vibration for Bridge 1 is a maximum of 0.047 m and an RMS displacement of 0.034 m. Accordingly, both bridges can still be subjected to a VIV that exceeds the allowable limit even though the actual damping ratios are taken into consideration.

In order to demonstrate the interactive behavior between the two bridges, the time histories of the vertical displacement in both bridges were compared one another at some intermediate wind velocities during a VIV. At a wind velocity of 11.7 m/s in Fig. 11(b), Bridge 2 reaches its maximum amplitude and, according to Seo et al. (2013), alternating “Ω” and “O” streamlines are generated between the two decks with the lock-in frequency tuned to Bridge 2. As shown in Fig. 12, Bridge 1 also shows a small vibration which is tuned to the frequency of Bridge 2. Fig. 12(c) clearly shows that the alternating large vortices also influence the motion of the downstream deck.

At a wind velocity of 12.5 m/s in Fig. 11(b), the amplitude of Bridge 2 decreases, while that of Bridge 1 increases, as shown in Fig. 13. The spectral density of Bridge 1 shows almost equal contribution of the two frequency components of each bridge, as shown in Fig. 13(c). This illustrates that the effect of a huge alternating vortex of Bridge 2 becomes weaker and Bridge 1 now shows a beat phenomenon in Fig. 13(a) owing to the closely spaced two frequency components of motion. A beat phenomenon was also reported by Kimura et al. (2008). As the wind velocity increases to 13.3 m/s in Fig. 11(b), the amplitude of Bridge 1 prevails with its own natural frequency, while the vibration of Bridge 2 becomes weaker, as shown in Fig. 14. However, the motional frequency of Bridge 2 remains unchanged from its natural

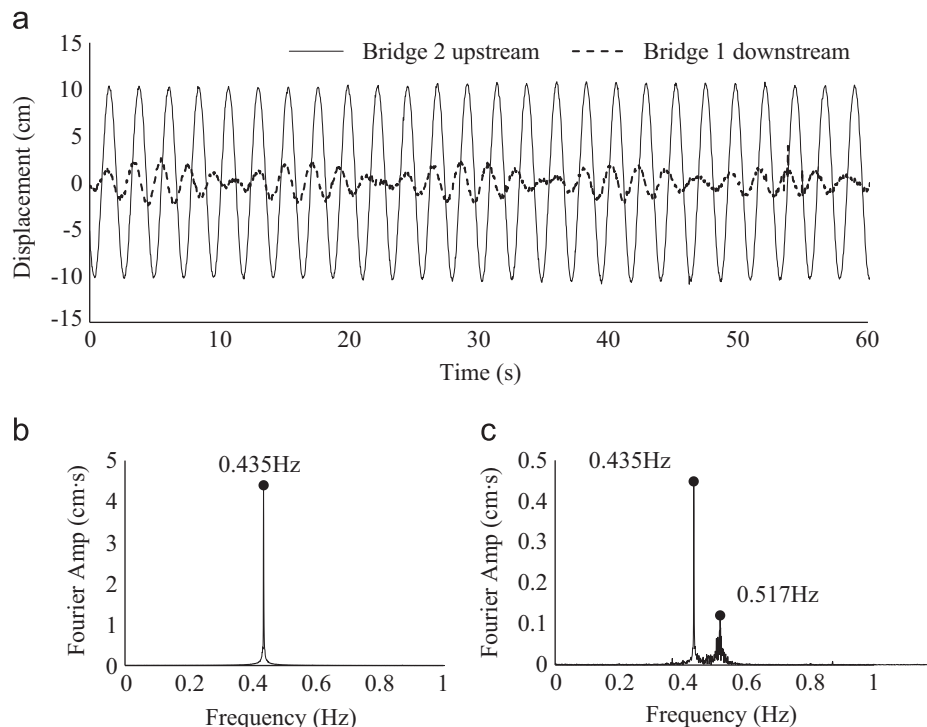


Fig. 12. Vertical vibration at the wind velocity of 11.7 m/s for NW wind: (a) displacement time histories for both bridges, (b) Fourier spectrum for Bridge 2 and (c) Fourier spectrum for Bridge 1.

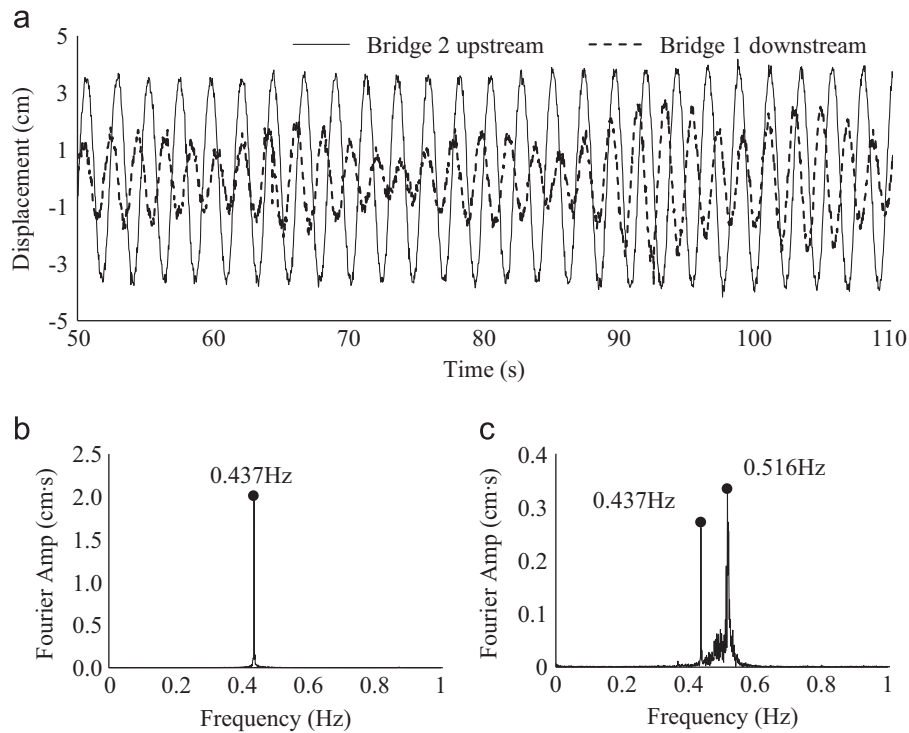


Fig. 13. Vertical vibration at the wind velocity of 12.5 m/s for NW wind: (a) displacement time histories for both bridges, (b) Fourier spectrum for Bridge 2 and (c) Fourier spectrum for Bridge 1.

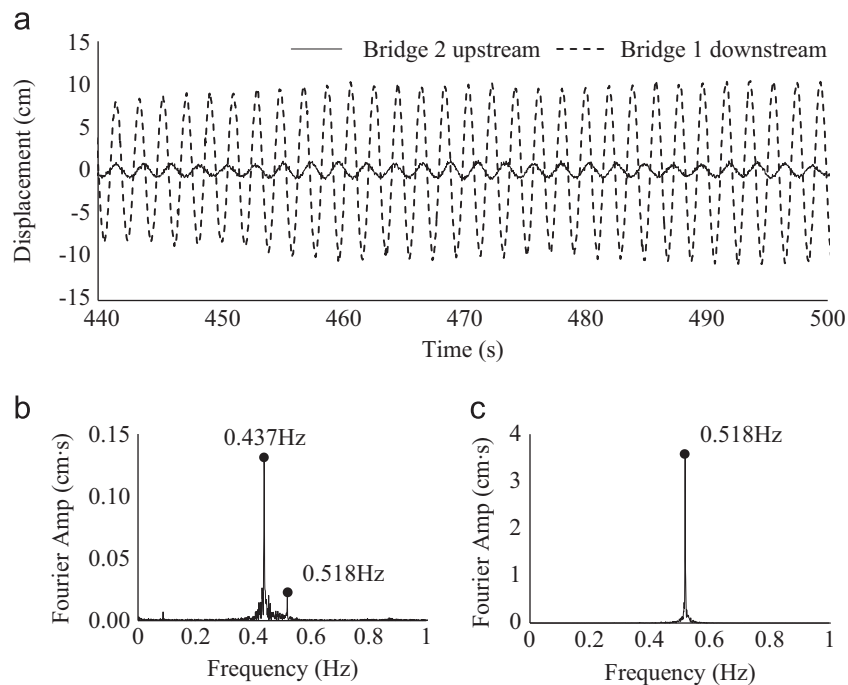


Fig. 14. Vertical vibration at the wind velocity of 13.3 m/s for NW wind: (a) displacement time histories for both bridges, (b) Fourier spectrum for Bridge 2 and (c) Fourier spectrum for Bridge 1.

frequency. In fact, the motional frequency of the upstream Bridge 2 is kept to the natural frequency of Bridge 2 in spite of the change in wind velocity.

3.4. Interactive behavior for SE wind

For the case of a SE wind, Bridge 1 is located upstream and has a higher vertical frequency than that of Bridge 2. The A–V curves for the low damping case in Fig. 15(a) show that the VIV is initiated

downstream in Bridge 2 at a wind velocity of 9.3 m/s. The amplitude of vibration increases with increasing wind velocity until the RMS amplitude reaches 0.223 m. At this point, Bridge 1 begins to vibrate upstream and, as a result, the motion of Bridge 2 is suddenly reduced. The VIV in Bridge 1 is developed freely upstream up to a wind velocity of 14.5 m/s, while the vibration in Bridge 2 is restrained.

Fig. 15(b) shows the case of the actual damping setup. Bridge 1 remained silent for all considered wind velocities owing to the

relatively high mechanical damping ratio. On the contrary, Bridge 2 is relatively free of VIV even downstream, showing maximum single amplitude of 0.08 m in RMS value.

Consequently, when considering the actual damping ratios of both bridges, Bridge 2 shows almost the same level of VIVs both for NW and SE winds while Bridge 1 shows a VIV downstream only for a NW wind downstream.

4. Operational monitoring of VIV in several wind conditions

The results of the wind tunnel tests indicate that the interactive behavior between the two decks depends on the wind direction and velocity as well as the mechanical damping ratios of the bridges.

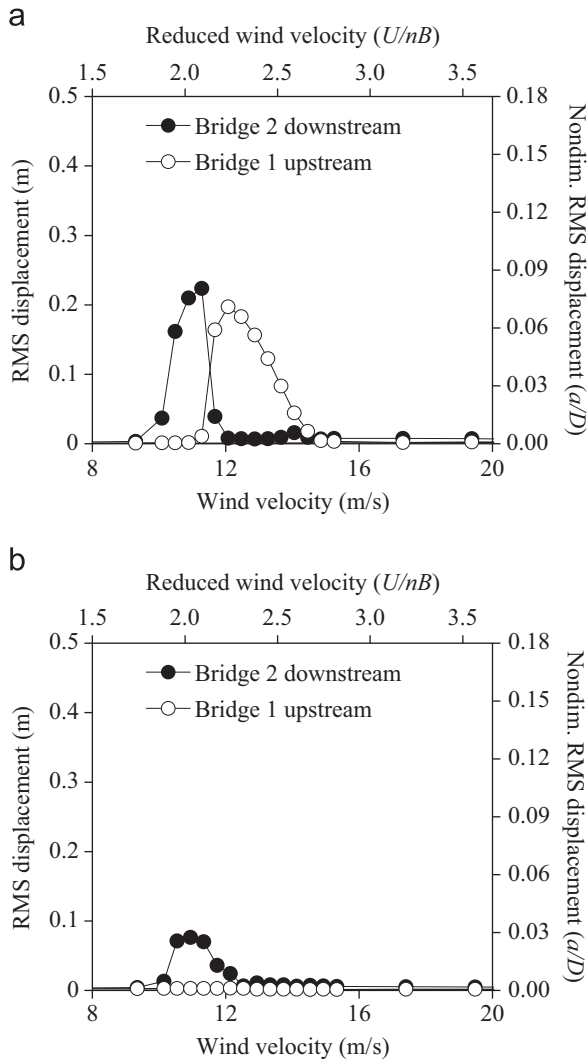


Fig. 15. A–V curves for a SE wind with the setup of: (a) low damping ratios and (b) actual damping ratios.

Table 2
Classification of wind data utilized according to the wind velocity and direction

Case	Classification	Date	Wind velocity (m/s)	Wind direction	Type of vibration Bridge 2	Type of vibration Bridge 1
1	Daily low wind (< 10 m/s)	2012/10/15, 10:00 to 2012/10/17, 09:20 and 19:00 to 0:00	~9.83	SE, SW, NW	TIV	No vibration
2	2011, strong VIV	2011/04/19, 14:00 to 16:10	9.59~11.85	NW	VIV	VIV
3	Daily high wind (≥ 10 m/s)	2012/10/17, 09:20 to 19:00	10.05~13.03	NW	VIV with TIV	VIV
4	Typhoon Tembin	2012/08/30, 09:00 to 16:00	~25.91	NW	VIV, TIV and buffeting	VIV and buffeting

Since both bridges have been in operation for several years, a lengthy accumulation of data sets measured from the built-in monitoring system are available. Several representative data sets were analyzed to examine how the observed interactive behaviors in the wind tunnel correspond to data for the actual bridges.

The older Bridge 1 has been closed to normal traffic for the past few years for upgrading load carrying capacity. Accordingly, the main source of vibration in Bridge 1 is the wind. Most vehicles use the newer Bridge 2 but the volume of traffic is not that high in such a remote place. Since the bridges connect two hill areas, the wind direction for achieving a meaningful velocity is always perpendicular to the spans. The dominant wind direction is NW in which Bridge 2 is located upstream.

Four sets of monitoring data were selected and are summarized in Table 2. Case 1 is for the traffic induced vibration (TIV) with a low wind velocity. Case 2 represents a strong VIV in Bridge 2 in 2011. This vibration is also the motivation for conducting this study. The data for case 3 were obtained from the three-day monitoring data introduced in Section 2.2. Among the daily monitoring data, only the parts involving high wind velocity were selected. The lasting time of the potential lock-in velocity was relatively shorter (about 3 min) than that for case 2 (20~30 min), so that the magnitude of the VIV is much less than that for case 2. Case 4 represents a typhoon and it is a unique case where the lock-in velocity for Bridge 2 is exceeded. The data for cases 1 and 3 were already utilized in evaluating the damping ratios of both bridges in Section 2.

The interactive behavior between the two bridges were investigated from the responses for cases 2, 3 and 4 which enable a qualitative as well as a quantitative comparison of the actual vibrations in field with those of the section model tests.

4.1. The case 2: strong steady-state VIV in Bridge 2

This is the case for the strong VIV in Bridge 2 shown in Fig. 2. The wind velocity was varying in a narrow range of 9~12 m/s. The wind data was collected by means of the anemometer on Bridge 1 downstream because of a temporal malfunction of the upstream anemometer. Based on Particle Image Velocimetry data reported by Seo et al. (2013), the wind velocity at the anemometer position appeared to decrease by about 10% compared to the free upstream during interference VIV in the upstream deck. The acceleration response at the center of the main span reached 150 gal, as shown in Fig. 2(b). Seo et al. (2013) reported this vibration in detail but only focused on the motion of Bridge 2. However, according to the wind tunnel tests, as shown in Fig. 12(a), a certain amount of vibration is expected, even in Bridge 1 during the maximum VIV. Accordingly, the interactive behaviors between the two bridges were examined in the monitoring data with emphasis on Bridge 1.

As shown in Fig. 2, the amplitude of the VIV is dependent on the time interval of the lock-in wind velocity. A 1-min period of a lock-in wind velocity of 12~13 m/s contributed about a 10 gal increment, as a rough estimation. The case shown in Fig. 2 also shows that a 20~30-min continuation of lock-in wind velocity is required to reach the steady state peak amplitude in Bridge 2.

Two parts of the data marked as 'INC' and 'MAX' in Fig. 2, represent the data for an increasing VIV and the maximum VIV, respectively. Fig. 16 shows the responses for the 'INC' portion. The amplitude of vibrations in Bridge 2 gradually increases at a lock-in wind velocity of 11.4 m/s. The large vortex streams are being developed behind Bridge 2 owing to the interference effect between two parallel decks (Seo et al., 2013). The acceleration response of Bridge 1 shows two spectral peaks in Fig. 16(e) at the first vertical frequencies of both bridges. Since the two frequencies (0.436 Hz for Bridge 2 and 0.513 Hz for Bridge 1) are close to each other, Bridge 1 shows a beat phenomenon in Fig. 16(b) which was also demonstrated in wind tunnel tests.

Fig. 17 shows the responses for the 'MAX' portion at which Bridge 2 shows steady-state maximum VIV. The dominant frequency of Bridge 1 is tuned to the frequency of Bridge 2, as shown in Fig. 17(e). This implies that Bridge 1 is now excited by the alternating interference VIV, which is in phase with the natural frequency of Bridge 2. This phenomenon was also demonstrated in the wind tunnel tests as shown in Fig. 12.

In summary, the field monitored data in Figs. 16 and 17 show good agreement with the wind tunnel responses in Figs. 13 and 12, respectively, in the sense that not only the triggering wind velocities are similar but also the motional patterns in both bridges are close in terms of the frequency components and the amplitude ratios between the two bridges.

4.2. The case 3: VIV+TIV in Bridge 2 and VIV in Bridge 1 during daily wind around lock-in velocity

This case includes the daily data for a relatively high wind velocity exceeding 10 m/s. Bridge 2 is subjected to excitations due to wind and crossing vehicles. As shown in Figs. 18(a) and (c), the raw acceleration in Bridge 2 shows high frequency components owing to crossing vehicles. However, Bridge 2 is mainly excited due to VIV since the filtered acceleration to the first frequency is almost similar to the raw data. The Fourier amplitudes in Fig. 18(c)

and (d) also demonstrate the dominant frequencies of the motions. Even though winds are controlling factors in the motion of Bridge 2, the lasting time of lock-in wind velocity is limited and the amplitude of VIV remains at the level of TIV.

Fig. 18(b) shows the VIV for Bridge 1. The overall level of vibration is smaller than that of Bridge 2 but the magnitudes of the maximum amplitude are almost the same in both bridges. It is interesting that both bridges experience interference VIV, even though this is hard to perceive, due to the parallel disposition of the structures, even in daily wind conditions.

4.3. The case 4: VIVs during the typhoon Tembin

The bridges are also affected by non-ordinary wind conditions, particularly during the typhoon season, as represented by case 4. Strong winds from the NW direction were observed during typhoon Tembin in August 2012. The maximum wind velocity in terms of a 10-min average was estimated to be 25.4 m/s. This wind case provides a special situation in which the wind velocity exceeded the lock-in velocity for Bridge 2 and, therefore, the VIV phenomenon possibly moved to Bridge 1 when the wind tunnel test results in Fig. 11(b) are considered.

Fig. 19 shows the V–A curves at the center of main span during the typhoon. Limited amplitudes of VIV were observed in Bridge 2 for a wind velocity of 12–13 m/s. However, the amplitudes of Bridge 1 downstream exceeded those of Bridge 2 when the wind velocity increased to 13–15 m/s. Fig. 20(a) compares the acceleration time histories for both bridges demonstrating the larger responses in Bridge 1 at a mean wind velocity of 14.8 m/s. Bridge 2 shows multiple peaks in Fig. 20(b) while Bridge 1 presents a VIV with a dominant peak in Fig. 20(c). Accordingly, the interactive behavior of larger vibrations in the downstream deck was successfully observed at a higher wind velocity over the lock-in velocity of the upstream deck. The upstream deck also showed a certain level of vibration with contributions of higher modes, probably due to

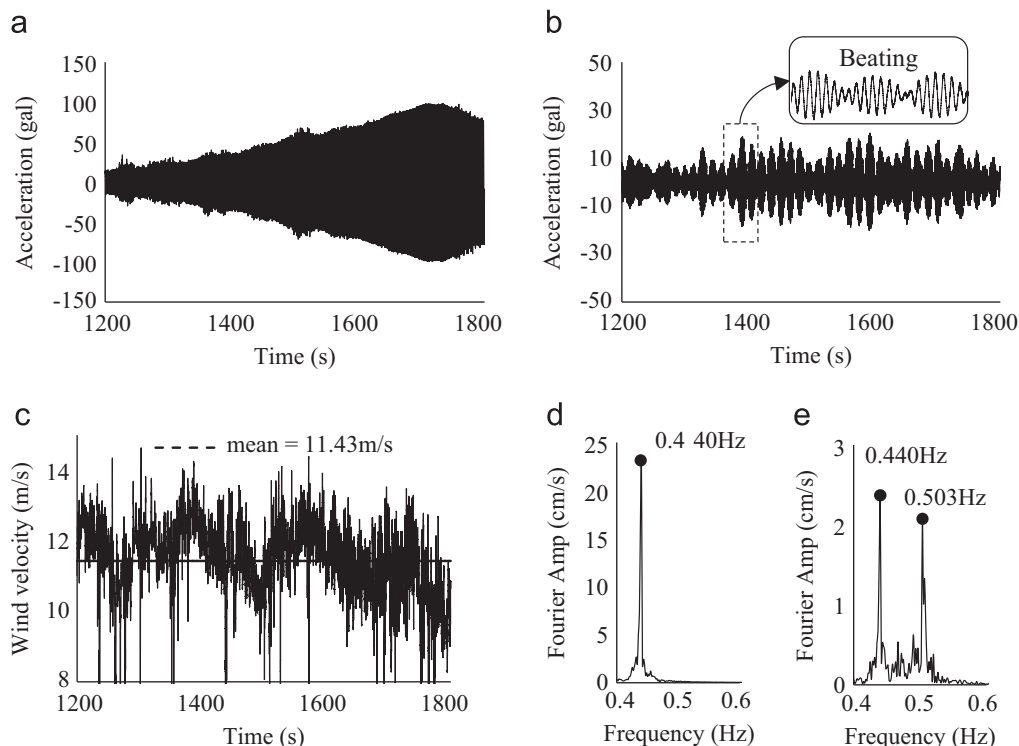


Fig. 16. Measured data at the center of the main span for the 'INC' portion: (a) vertical acceleration of Bridge 2, (b) vertical acceleration of Bridge 1, (c) wind velocity, (d) Fourier spectrum of (a) and (e) Fourier spectrum of (b).

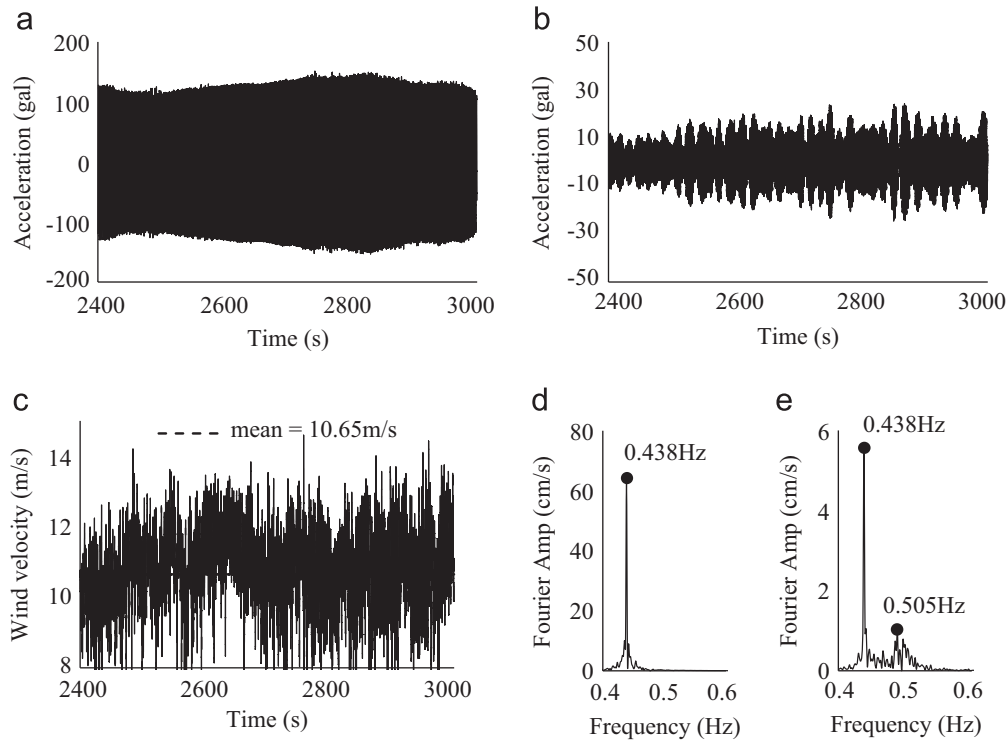


Fig. 17. Measured data at the center of the main span for the 'MAX' portion: (a) vertical acceleration of Bridge 2, (b) vertical acceleration of Bridge 1, (c) wind velocity, (d) Fourier spectrum of (a) and (e) Fourier spectrum of (b).

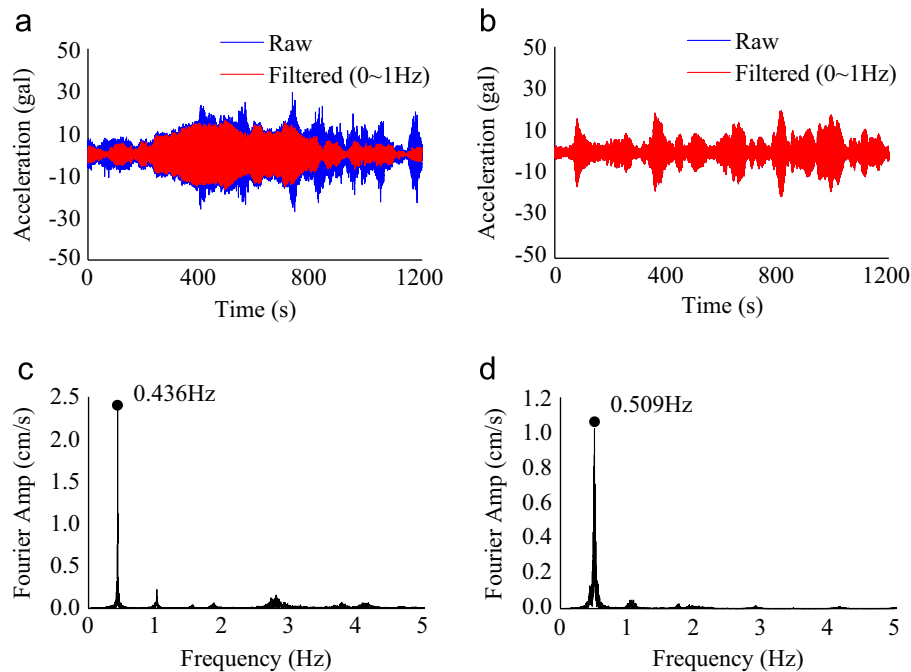


Fig. 18. Measured vertical acceleration data at the center of the main span for daily lock-in wind velocity (mean wind velocity = 11.76 m/s): (a) VIV + TIV in Bridge 2, (b) VIV in Bridge 1, (c) Fourier spectrum of (a) and (d) Fourier spectrum of (b).

the turbulence effect of upcoming winds as well as crossing vehicles.

4.4. Summary of operational monitoring of VIV

The four cases of monitoring data are shown together in Fig. 21. As shown in Fig. 21(a), the VIV in 2011 was extraordinary large

compared to the other observations. The duration of the lock-in wind velocity continued repeatedly for hours and the amplitude of the VIV increased to the nearly maximum level, which corresponds to the steady-state amplitude identified in wind tunnel tests.

For a better assessment of ordinary interactive behaviors, the data for the VIV in 2011 were excluded and shown again in Fig. 22.

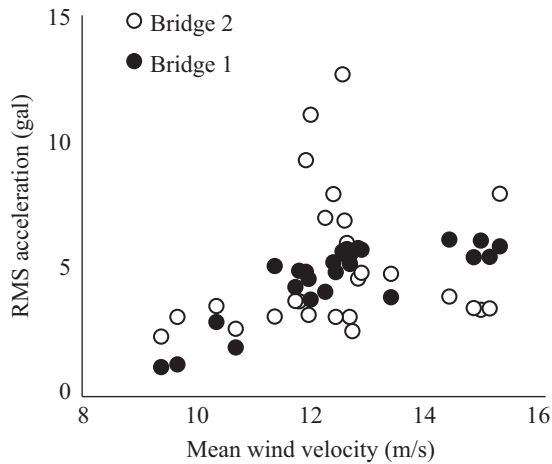


Fig. 19. A–V curves for both bridges during typhoon Tembin.

For the range of low wind velocity, Bridge 2 was subjected to a TIV while Bridge 1 was silent, due to the fact that it was closed to traffic. When the wind velocity reached 12~14 m/s, interference VIVs and the lock-in phenomenon were identified in Bridge 2. The VIV upstream from Bridge 2 also induced VIVs in Bridge 1 as shown in Fig. 22(b). When the wind velocity increased to over 14 m/s, Bridge 1 was subjected to a VIV and a lock-in phenomenon was identified up to a wind velocity of 18 m/s. For a high wind velocity of over 20 m/s, both bridges were subjected to buffeting vibrations.

5. Conclusions

The interference vortex-induced vibration (VIV) between two parallel cable-stayed bridges was investigated with a particular focus on the mutual interactive behaviors in both bridges as a function of wind velocity and direction. Both decks were elastically

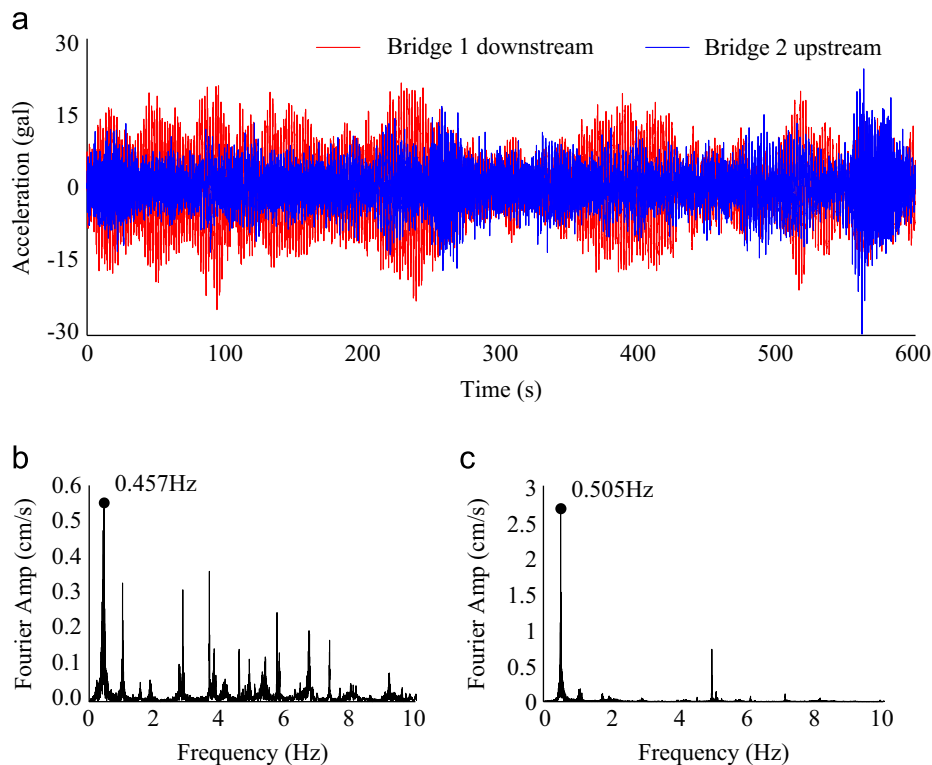


Fig. 20. Measured vertical acceleration data at the center of the main span for the typhoon Tembin (mean wind velocity = 14.8 m/s): (a) time histories for Bridge 2 and Bridge 1, (b) Fourier spectrum for Bridge 2 and (c) Fourier spectrum for Bridge 1.

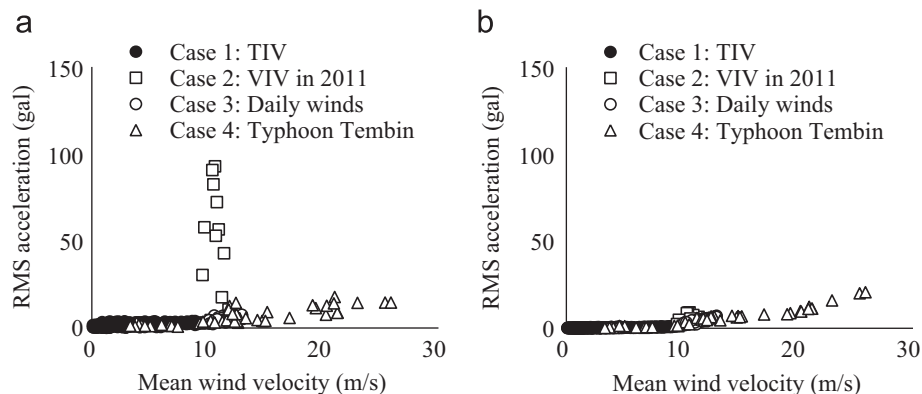


Fig. 21. RMS vertical acceleration from operational monitoring: (a) Bridge 2 and (b) Bridge 1.

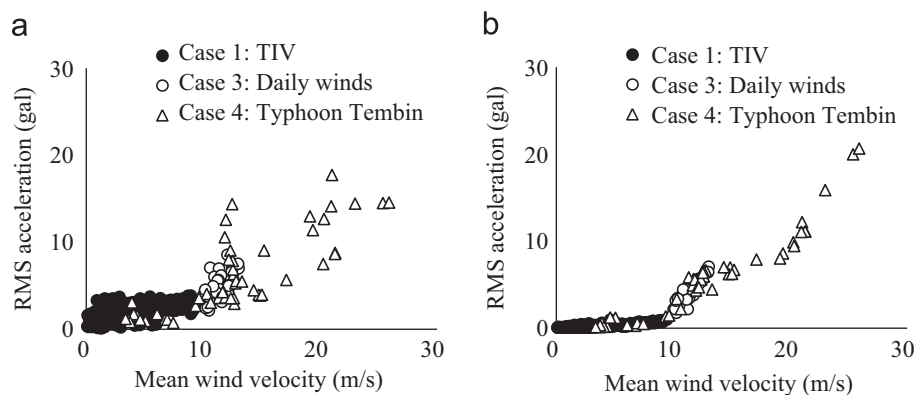


Fig. 22. RMS vertical acceleration except for the huge VIV in 2011: (a) Bridge 2 and (b) Bridge 1.

supported on springs with the setting of the mechanical damping ratios identified from an operational modal analysis for the actual bridges. The mechanical damping ratio of Bridge 2, which was subjected to a significant VIV, was estimated to be 0.28% while that of Bridge 1 was 0.58%. Accordingly, even though the two bridges adopt the same type of structural system with girders having a similar cross-sectional shape, the mechanical damping ratios are different. Even though perceivable vibrations were reported only for Bridge 2, the wind tunnel tests demonstrate that VIV can be induced, not only in Bridge 2 but also in Bridge 1. In fact, if the mechanical damping ratios were the same in both bridges, the maximum amplitude of the VIV in Bridge 1 would likely exceed that of Bridge 2. However, owing to the higher damping ratio of Bridge 1, the amplitude level of the bridge was lower than that of Bridge 2.

Another important factor regarding the interaction behavior is the wind direction. Based on the wind tunnel tests, Bridge 2 shows almost the same level of VIV both for northwest (NW) and southeast (SE) winds while Bridge 1 shows a VIV only for a downstream NW wind when considering the actual mechanical damping ratios of the bridges. Since the main wind direction is NW for meaningful wind velocities at the bridge site, Bridge 1 may show a VIV downstream. However, this phenomenon would be observed only for higher wind velocities that exceed the lock-in of Bridge 2 upstream. This is one of the possible reasons for why a VIV has seldom been reported for Bridge 1.

It should also be noted that the initiation of VIV in the upstream bridge is relatively free from the motion of the bridge downstream. However, the motion of the bridge downstream is significantly affected by the motion of the bridge upstream. When the natural frequency of the upstream bridge is lower than that of downstream bridge, which is the case for the Jindo Bridge subjected to NW winds, the bridge upstream vibrates first. As the wind velocity increases, the VIV moves to the bridge downstream. During wind velocities in transition, the deck downstream vibrates with two dominant frequencies, i.e., one for its own frequency and the other for the frequency of the upstream bridge. Accordingly, the motion of the deck downstream shows a beat phenomenon.

The interactive behaviors observed in a wind tunnel were also examined through field monitoring by using the built-in sensors in both bridges. The overall interactive behaviors are in good agreement to those in a wind tunnel in terms of the threshold wind velocity, the frequency components of motions and the amplitude ratio between two bridges. However, even over the lock-in wind

velocity, the upstream deck still shows a certain level of vibration which, in turn, slightly weakens the tuned VIV in the downstream deck when compared with experimental results. The vibration in the upstream deck seems to be the effect of fluctuations in the wind velocity as well as crossing vehicles which were not considered in the wind tunnel tests. However, the overall behaviors of the actual bridges conform to the observations made in the wind tunnel. A beat phenomena at the downstream deck was also identified in the monitoring data. Unfortunately, a strong wind was not observed in the SE direction and, as a result, it was not possible to evaluate interactive behavior for this situation.

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